Banavathu Mahidhar Naik

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	7.00
narayana junior college ,vijayawada	Class XII (board of intermediate education , andhra pradesh, india)	2024	98.00%
geethanjali high school , vinukonda	Class X (board of secondary education , andhra pradesh, india)	2022	95.83%

SCHOLASTIC ACHIEVEMENTS

- Secured 360th rank in the **Srinivasa Ramanujan Maths Competition** conducted by IIT Madras.

SKILLS

- **Programming Languages:** C++, C, Python, Javascript , SQL , Assembly.
- **Databases:** MongoDB, PostgreSQL.
- **Web Technologies :** HTML, CSS, ReactJS.
- **Operating systems:** Windows , Ubuntu.
- **Tools:** Git ,Github,SolidEdge ,Visual Studio Code , LaTeX.

RELEVANT COURSES

- **Computer Science:** Introduction to Programming, Operating Systems, Database Management Systems, Theory of Computation, Algorithms, Data Structures, Computer Architecture, Software Developement Fundamentals, Discrete Mathematics, Introduction to Compilers.
- **Mathematics:** Linear Algebra, Probability & Random Variables, Calculus, Transfrom Techniques, Introduction to Statistics.
- **Others:** Communication Skills, Digital Fabrication, Introduction to Artificial Intelligence.

PROJECTS

Experience with xv6 Operating Systems | C ,RISC-V

- Extended the xv6 kernel by implementing **custom system calls** and per-process system call accounting, improving kernel functionality and enabling detailed monitoring of system call usage.
- Replaced the default **Round Robin scheduler** with a **Multi-Level Feedback Queue (MLFQ) scheduler** supporting **priority-based preemption**, **configurable time slices**, and **global priority boosting** for starvation prevention.
- Built a **virtual memory subsystem** using **page-fault-driven lazy allocation**, the **Clock page replacement policy**, and a **disk-backed swap space** with **swap-in and swap-out support**, enabling processes to execute beyond memory limits.
- Developed **configurable disk scheduling** for swap I/O, integrated **RAID (0/1/5)-based multi-disk swap storage** for **disk failure recovery**, and implemented **kernel statistics** to analyze disk and swap activity across different workloads.

Database Access Methods and Query Plan Analysis | PostgreSQL

- Analyzed **query runtime** across different access methods and indexing strategies, comparing **sequential scans, B-tree indexes, clustered indexes, index-only scans** to understand their impact on query performance and selectivity.
- Analyzed PostgreSQL query plans at **compile time and runtime,** comparing physical operator trees and execution costs to **detect plan switches** across varying query parameters; evaluated optimizer decisions as early,delayed, or correct and **identified optimal switch points** based on actual query performance.

Computer Networks -- Secure Echo Client/Server | C, TCP Sockets, OpenSSL, Linux

- Implemented **dual-stack IPv4/IPv6 TCP connections** with dynamic hostname resolution, validated using **Wireshark and tcpdump** packet capture.
- Scaled to **multiple concurrent clients** via a **select()-based I/O multiplexing** architecture with real-time timestamped logging for every client message.
- Secured end-to-end communication using **OpenSSL TLS with CA certificate** verification and a custom application-level **authentication** protocol ensuring credentials are never transmitted in plaintext.

WeatherCast - Smart Weather Dashboard | Node.js, MongoDB

- Built a full-stack weather app using **React** and **TypeScrip**t with JWT-based secure login. Weather data is fetched from **Open-Meteo REST API** through an **Node.js** backend.
- Added an interactive map using **Leaflet map** where users can click anywhere to get weather for that location. City search with live suggestions, favorite locations, and search history are all stored in **MongoDB Atlas**.
- Created a weather dashboard using **Recharts** showing hourly temperature, rain chance, and UV index graphs with a 7-day forecast. Automatic weather warnings like heat alerts and storm warnings are generated from live **API data**.

EXTRACURRICULAR

- **Codeforces** | Solved 300+ competitive programming challanges and achieved a **peak rating of 1314**(Pupil rank).
- **CodeChef** | Attained a **peak rating of 1544** by solving algorithmic problems and optimizing data structures.
- Member in National Service Scheme.